11C — Deep-Dive Feature Audit: Human Foundation, Psychology, Social Graph & Karma (Part 3 of 5)

Dynamic Psychology: Inverted Immortal Hunger Formula (+6%/Stage), 5 Frozen h.state Emotions, Broken values8 Fulfillment Loops, Paradoxical heartDemon Enlightenment & Missing Qi-Deviation (走火入魔) Climax

- **Target Codebase**: JianghuSect 2 (JS2-v2.6.0 in Google Drive Js2 new)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **Immutable Rule**: **NEVER modify js/human_core.js** (155:2f9c9dd7). All fixes operate in js/ai.js, js/cognition.js, js/sect.js, js/combat.js, and js/cultivation.js.

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1. [HUM-PSY-01] Inverted Metabolic Hunger Bug (js/ai.js:972): Higher-Stage Cultivators Starve 66% FASTER Than Mortals!

1.1 Verbatim Code Proof (js/ai.js:972 vs 979)

```javascript
met: 0.82 + (100 - (E(n, 'physiology.vitality') | 0)) / 300 + (cStage ? 0.06 * cStage : 0),
...
restDrainMult: eqStage > 0 ? (1 / (1 + eqStage * 0.75 + Math.pow(Math.max(0, eqStage - 3), 1.5) * 0.45)) : 1.0
```

- **Direct Algebraic Contradiction**:
  - While restDrainMult (line 979) properly reduces sleep need as eqStage rises, **met (line 972, multiplying s.satiety loss every minute) ADDS +0.06 * cStage!**
  - A Stage-9 Immortal (cStage = 9) suffers +0.54 extra metabolism (1.36 vs 0.82 — **starving 66% faster than an unawakened mortal peasant!**), forcing

2,000-year-old ancestors to sprint to the Canteen 4+ times a day instead of practicing Xianxia **Bigu (辟谷 — Inedia)**!

- **Exact Fix**: Replace `+ (cStage ? 0.06 * cStage : 0)` with stage-based **Bigu attenuation**: `const biguMult = (c && ['BODY', 'DEVOURING_FLESH', 'ARENA_GLORY'].includes(c.path)) ? 1.05 : 1 / (1 + eqStage * 0.32);` so Stage-9 Qi/Dao Immortals burn `~0.26×` mortal food while Body-Refiners retain hearty appetites!

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2. `[HUM-PSY-02]` Five Frozen `h.state` Psychological Dimensions (`comfort`, `pain`, `confidence`, `fear`, `anger`) & Unbounded `n.grief` (`js/ai.js`, `js/combat.js`)!

`human_core.js:289` creates 16 rich dynamic state dimensions, yet **5 of them are dead or half-dead in live simulation**:

1. **`h.state.comfort` (`Frozen at 60 forever`)**: `0` writes anywhere in the codebase! Sleeping outdoors on bare ground (`ctx.out = true`), working in storms/blizzards without weather immunity, or resting in upgraded `Lv.4–5` dorms/tea gardens never updates `comfort` or feeds `mood`.
2. **`h.state.pain` (`Frozen at 0 forever`)**: Even with three `sev: 3` broken ribs/skull injuries (`n.inj`), `AI.injure` (`ai.js:814`) never sets `pain`, and `A.heal` / pain-relief pills never relieve it! Sync `s.pain` dynamically with active `n.inj` severity (`and combat wounds`), letting `painTolerance` mitigate `stress`/`focus` loss from `pain`!
3. **`h.state.confidence` (`Frozen at 50 forever`)**: Winning duels/tournaments/breakthroughs never raises `confidence`, and repeated defeats/failed breakthroughs never lower it (`nor feed` chooseSkill/breakthrough `hesitation`).
4. **`h.state.fear` & `h.state.anger`**: Decayed every tick in `AI.needs` (`line 1007`), yet `combat.js` never spikes `fear` when outmatched by a `+2` stage Boss or watching an ally die, and `anger` (`only +6 in argue`) never escalates `grudge` initiation or reckless `aggression` in combat!
5. **`n.grief` (`ai.js:849`)**: Increments `+4..+8` whenever a friend/companion dies, **never decays**, and is **never read**! Decay `n.grief` during `A.pray` at `ancestral` (`mourning at the fallen loved one's spirit tablet!`) and let active `grief > 0` gently pull `n` toward `A.pray` while dampening `mood`.

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3. [HUM-PSY-03] Four Broken values8 Fulfillment Loops (h.fulfil) & Ignored A.eat.vf Property (js/ai.js:301, 995–1002)!

Every h.fulfil[k] decays by -0.0042/tick (-6.05/day), driving unreplenished values to 0 (maximum psychological deprivation). Tracing all AI.vf(...) calls exposes **four starved life values**:

1. **Ignored Static vf: { familyClan: 0.1 } on A.eat (line 301) & Zero familyClan Gain in pair (lines 726–745)**:
   - A.eat declares vf: { familyClan: 0.1 }, **which AI.step NEVER reads**! Worse, **AI.pairTick (chat, date, teach with spouse, children, parents, or sworn friends!) calls AI.vf ZERO TIMES**! Even a devoted spouse (values.familyClan = 95) who dates their partner daily sits at fulfil.familyClan ≈ 0!
2. **wealthResource Only Replenished by A.mine (Digging Rocks!)**:
   - Earning Spirit Stones via outsideTrade, MERCANTILE_DAO contracts, or CPE.mint donations (> 30 CP) never calls AI.vf(n, { wealthResource })—causing rich merchants and treasurers to feel perpetually impoverished (fulfil.wealthResource = 0) and gravitate toward manual mining!
3. **powerStatus & honorLegacy Starved Outside Bureaucratic Desk Duty**:
   - Replenish powerStatus & honorLegacy on tournament/duel victories, Peak Master/Elder command, raid defense victories, and teach (mentoring disciples)!

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4. [HUM-PSY-04] Paradoxical heartDemon Enlightenment Bonus (js/ai.js:440, 464), Missing Autonomous Counseling (AI.socialStart), & Missing 走火入魔 Qi-Deviation Climax!

4.1 High heartDemon Grants 4× MORE Peaceful Dao Enlightenment (js/ai.js:440 & 464–466)!

Inspect lines 440 and 464–466 verbatim:

```javascript
if (W.rng.chance(0.0016 * (1 + s.heartDemon / 30))) { AI.insight(W, n); }
```

```
...
AI.insight = (W, n) => {
  const kind = W.rng.next();
  if (kind < 0.8) { CULT.gain(n, 6 + W.rng.int(0, 8)); HC.experience(n, 'insight',
0.6); ... }
  else { n.h.state.heartDemon = clamp(n.h.state.heartDemon + 8, 0, 100); ... }
};
```

- **Logical Inversion**: At `heartDemon = 90`, trigger probability quadruples (`4.0×`), **and `80%` of triggers STILL award pure peaceful Dao Enlightenment (`+XP, +composure, +20 mood`)**! High Heart Demons accidentally function as a massive enlightenment buff!
- **Fix**: Make `const insightProb = clamp(0.82 - (s.heartDemon / 140) + (E(n, 'intellect.willpower') - 50) / 250, 0.15, 0.90);` so low-`heartDemon` cultivators achieve pure enlightenment while high-`heartDemon` cultivators face dangerous inner-demon trials (unless they triumph through `willpower`, yielding a rare Bodhi breakthrough!).

4.2 Zero Autonomous Heart-Demon Counseling in `AI.socialStart` & Missing `heartDemon >= 95` Qi-Deviation (走火入魔) Event

1. **Add `'counsel'` to `AI.socialStart` (`ai.js:768–795`)**:
   - When an NPC's apprentice, Taoist Companion, friend, or sectmate has `heartDemon >= 50` (or an active `SECT.HEART_DEMON` obsession) and `n` is a `BUDDHA_DHARMA` / `HEART_SOUL` / `SAGE_SUPPORT` practitioner, `mentor`, or `companion`, allow proposing a `'counsel'` pair session at `tea_garden` / `shrine` (`calling SECT.HEART_DEMON.counsel` and reducing `heartDemon` by `-12..22`)!
2. **Authentic `heartDemon >= 95` Qi-Deviation Crisis (走火入魔)**:
   - When a non-controlled cultivator lingers at `heartDemon >= 95` during `cultivate` or `breakthrough` and fails a `willpower + composure` check:
     - Trigger a real emergent **Qi-Deviation Crisis**: either **Meridian Backlash** (`sev: 2` internal injury, `-15%` stage progress, `heartDemon -> 55`), **Berserk Inner-Demon Bout** (non-lethal subdual duel where `discipline` / `guard` / `mentor` steps in to subdue and calm them!), or—if `integrity < 30 && loyalty < 35`—**Temptation Into Demonic Defection / Forbidden Art Unsealing**!
3. **Unify `n.h.mind.memories` & `n.cognition.memory` (`02A`) + Guard `killed_foe` Against Mindless Beasts**:
   - Only trigger `HC.experience(n, 'killed_foe')` (`empathy -0.9, aggression +0.4`) when slaying **humanoid cultivators**, never when hunting

wild boars/fish, so righteous gate guards don't turn into cold-blooded sociopaths from pest control!